Student Management System
Project Report

Names: Ntaganira kwitonda theos (224004187)

Ndayishimiye Irakoze Heritier (224004851)

1. Introduction

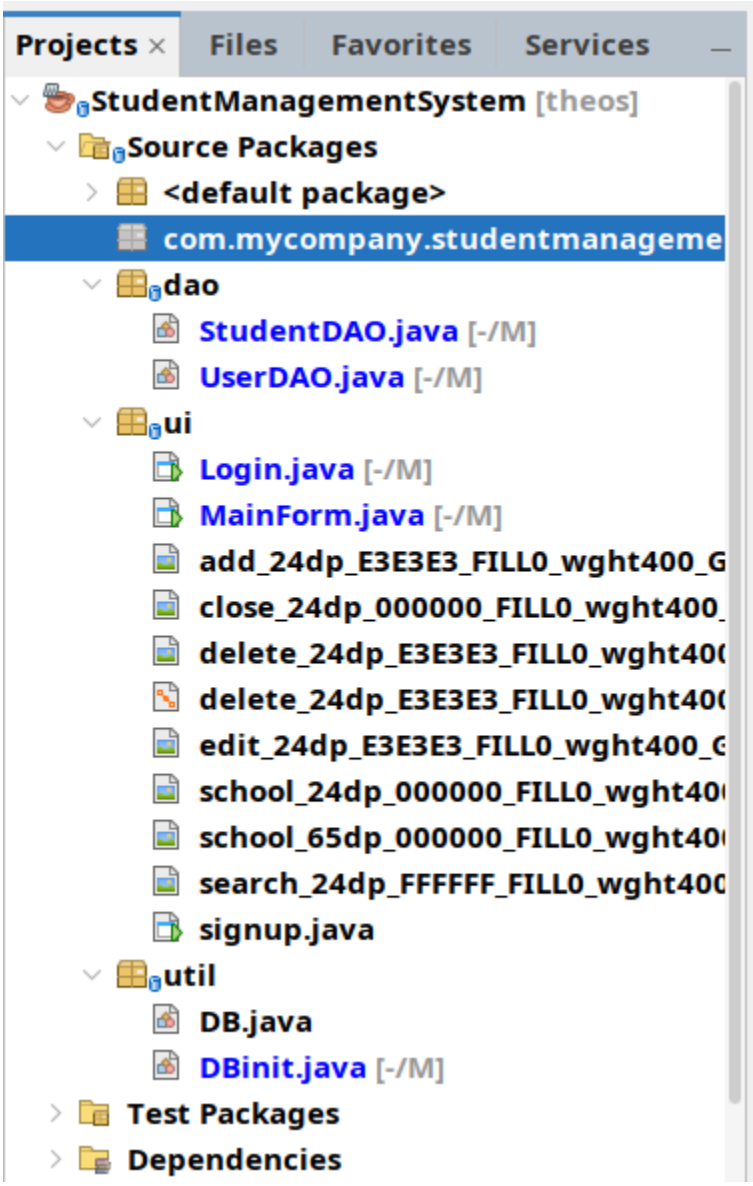
This Student Management System is a desktop Java application developed to manage students through GUI. The system provides a login page, sign up page, a main management page, SQLite database, and CRUD operations needed to add, update, delete, search, sort and display student information.

3. Development Environment and Tools

Tool / Technology	Use in the project
Java	Used to implement the application logic.
Java Swing	Used to create the UI and it's components
NetBeans IDE	IDE of choice used because SWING UI builder
SQLite	Lightweight local database used to store student records in sms.db
JDBC	Used to connect Java classes to the SQLite database and execute SQL commands.
Git and GitHub	Used for version control, branching, and project submission with 3 branches (hertier,theos,main).

4. Project Structure

The submitted project follows a package-based structure that separates the UI,util,data access object(DAO). This makes the project easier to read, maintain, and extend.



Main files and their roles:

- DB.java - opens the JDBC connection to the SQLite database.
- DBinit.java - contains creation and updating of database tables using queries
- StudentDAO.java - performs database operations such as create table, add, update, delete, search, and sort.
- UserDAO.java - performs login and register operations for lectures (users).Student.java - stores student attributes such as id, name, email, gender, course, and marks.
- LoginForm.java - provides the login screen with username, password, remember me, reset, login message, and register as a lecturer.
- MainForm.java - provides the main student management screen with form inputs, menu items, search, sort, and JTable display.
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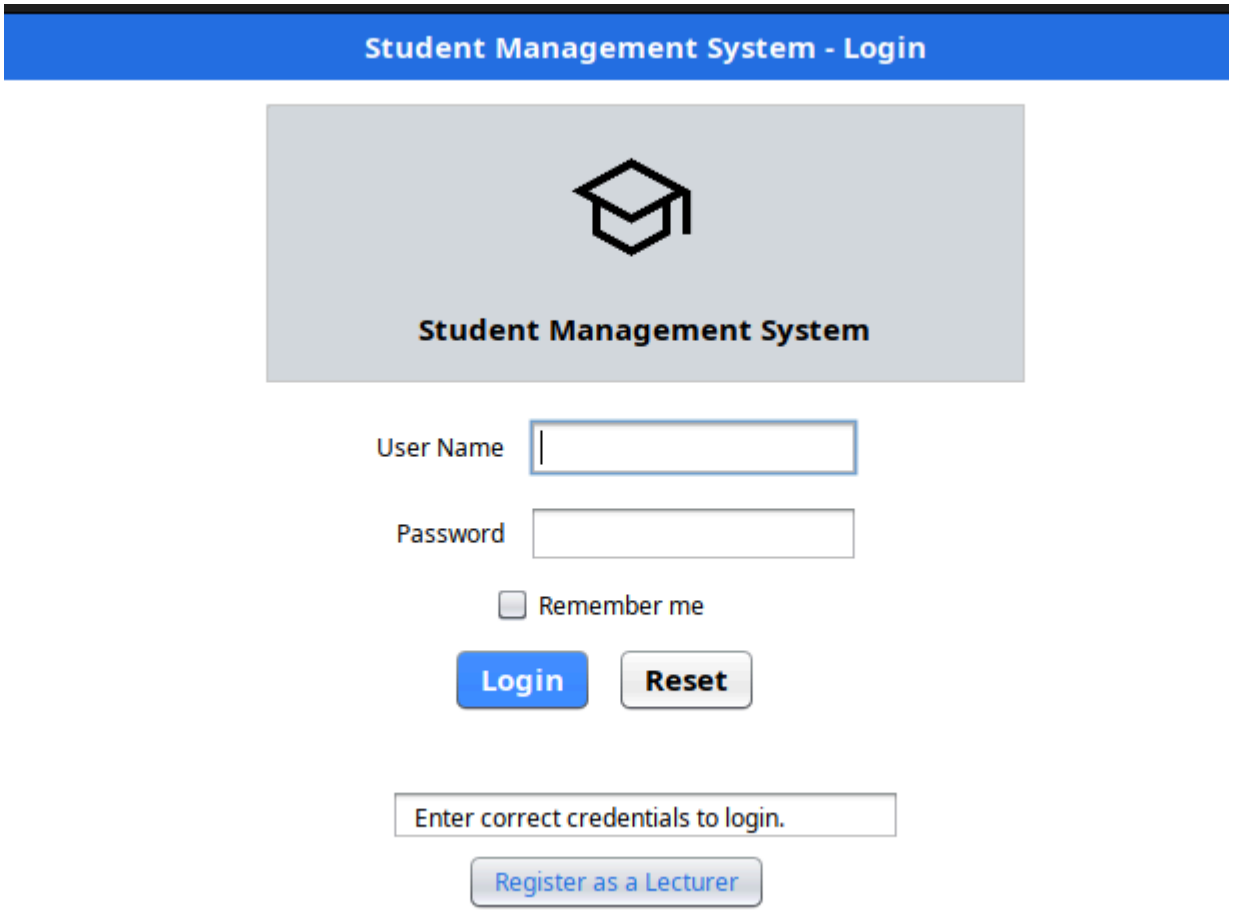
5. System Design and Functional Overview

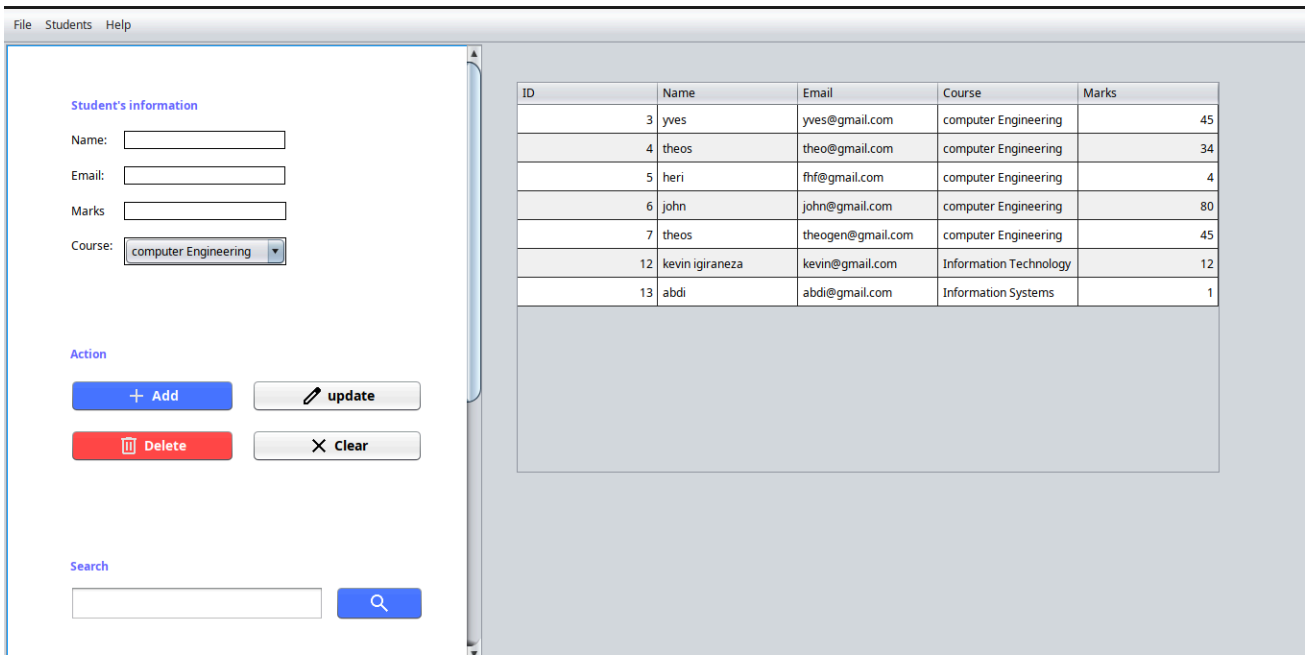
5.1 Login Form

The login form contains username and password fields, Login and Reset buttons, a Remember Me checkbox, register as a lecturer and a logo image.



5.2 Main Form

The main form is the central workspace for student management. It contains text fields for name, email, and marks, courses combo box, and search category; buttons for Add, Update, Delete, Clear, Search, and Show All; radio buttons for sorting; a table for record display; and a menu bar with File, Students, and Help menus and also filters.



6. Students Database Design

The application uses an SQLite database stored in the file students.db. The table is created automatically if it does not exist.

Field	Description
name	Student full name.
email	Student email address.
course	Selected course, for example computer engineering, IT, IS, or computer science.
marks	Numeric scores are stored as a real value.

7. Implemented Functionalities

Function	Current behavior	Status
Login	Checks username and password, and opens MainForm on success.	Implemented
Reset login	Clears login fields, resets checkbox, message label, and progress bar.	Implemented
Add student	Adds a new record after required field and marks validation.	Implemented
Update student	Updates the selected record after required field and marks validation.	Implemented

Delete student	Deletes the selected record after confirmation.	Implemented
Filter	Filter the students based on a selected course	Implemented
Search	Searches by ID, Name, Email, Gender, or Course using a combo box.	Implemented
Sort	Sorts student records by Name or Marks using radio buttons.	Implemented
Logout	Returns from MainForm to LoginForm and shows a success message from the file menu.	Implemented
Exit	Close the whole application from the File menu.	Implemented

9. Validation and Business Rules

- Add and Update operations check that the main input fields are not empty before saving changes.
- Marks must be a valid number.
- Marks cannot be less than 0.
- Marks cannot exceed 100.
- Searching by ID expects a numeric input and warns the user when a non-numeric value is entered.
- Can not add an email already in the database
- Name search also uses an exact case-insensitive comparison so typing KEVIN returns Kevin.

13. Conclusion

In conclusion, the submitted Student Management System successfully implements the main practical requirements of a desktop CRUD application. The project uses Java Swing for the user interface, SQLite with JDBC for persistent storage, and GitHub for collaboration. The application supports login, add, update, delete, search, sort, logout, and an about page.